

EuBIC-MS Annual report 2023-2024

The European Bioinformatics Community for mass spectrometry (EuBIC-MS) is the EuPA initiative for mass spectrometry (MS)-related bioinformatics. Its aim is to improve bioinformatics for MS-based research through the setup of community-driven dynamics, thereby improving collaboration, funding, publication, and training activities. Through its different activities, EuBIC-MS aims to raise awareness for the benefits of open science and continuously supports open software. It was started in November 2015 (Vaudel, 2016) to bring together the MS and bioinformatics communities in an open, collaborative, and constructive environment. The initiative is carried enthusiastically by the bioinformatics community and welcomes everyone willing to help.

EuBIC-MS organizes a dedicated yearly conference, bringing together researchers in the fields of mass spectrometry and bioinformatics, junior scientists and industry partners from across Europe. Every year, this conference alternates between a Winter School with keynotes and educational workshops, and a Developers Meeting consisting of collaborative project sessions with the participants (hackathons). EuBIC-MS also actively contributes to bioinformatics hubs, and provides various workshops and presentations during international MS and proteomics conferences, such as those of the American Society for Mass Spectrometry, the Human Proteome Organization, and EuPA. Additionally, EuBIC-MS coordinates various research projects conducted by the community.

In this document, we report our activities and achievements for the period between October 2023 and September 2024.

Winter Schools

The EuBIC-MS Winter School on computational MS is sponsored by the European Proteomics Association (EuPA) and several other parties, including but not limited to companies with interest in proteomics and mass spectrometry. The Winter School brings together scientists from both academia and industry to present and discuss their research in workshops, keynote lectures, flash talks and poster presentations. So far, we have successfully organized four Winter Schools: the first one in Semmering, Austria in 2017 (Willems et al., 2017), the second in Zakopane,

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Poland in 2019 (Kopczynski et al., 2019), the third in March 2022 at the Instituto Gulbenkian de Ciência in Oeiras (Portugal), and the most recent, fourth, one in Winterberg (Germany) the 15th-19th of January 2024. It was sponsored by BSi, MSAID, Evosep, Matrix Science and de.NBI.

Our Winter School took place from 15-19 January at Hotel Der Schöne Asten, Winterberg, Germany. It was a great success, with 116 participants from 15 different countries. The event was sponsored by BSI, EvoSep, MSAID (Gold sponsor) and Matrix Science (Bronze sponsor), and co-organised by PROTrEIN, EuPA and de.NBI.



The first day was dedicated to educational workshops organized by PROTrEIN, one invited and one sponsored workshop from MSAID. The participants could choose between three different full-day workshops arranged by PROTrEIN entitled “Introduction to mass spectrometry based proteomics workflows and data analysis” (Ayesha Feroz, Carlo De Nart, Chiara Sopegno, Louise Buur, Zoltan Udvardy), “Workflows for PTM identification and validation” (Aditi Sharma, Arslan Siraj, Arthur Grimaud, Pinar Altiner, Marc Pauper) and “Prediction models in proteomics” (Alireza Nameni, Mostafa Kalhor, Shamil Urazbaktin, Zahra Elhamraoui, Prajwal D’Souza); or two successive half-day workshops entitled “GitHub Actions Demystified: A Hands-On Workshop” (Pieter Verschaffelt, Ghent University, Belgium) and “AI-powered analysis of bottom-up proteomics data with CHIMERYs” (Dr. Martin Frejno, MSAID).

This first educational day was followed by three and a half days of keynotes (morning) and parallel workshops (afternoon). Unfortunately, one keynote speaker could not attend the event in person after missing his flight, hence his presentation was done online (and successfully, even though the WIFI was crap). In total, there were 11eleven keynote talks, including one sponsored one, given by Thomas Burger (CNRS, France –online–), Cecilia Lindskog (Uppsala University, Sweden), Ben Neely (NIST, United States of America), Nils Hoffman (FZ Jülich, Germany),

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Juan Antonio Vizcaino (EMBL-EBI, UK), Laurent Gatto (UCLouvain, Belgium), Wen-Feng Zeng (MPI Biochemistry, Germany), Dirk Valkenborg (Hasselt University, Belgium), Robbin Bouwmeester (Ghent University, Belgium), Niveda Sundararaman (Cedars-Sinai, USA) and Vadim Demichev (Charité University, Germany - EvoSep sponsored-).

EuBIC-MS invited all keynote speakers (or members of their team) to organize a workshop on any topic or tool that they deem relevant to the field. In 2024, the participants could choose among the following parallel workshops:

Tuesday:

- PRIDE and ProteomeXchange: Introduction, submission and re-usage of data (Juan Antonio Vizcaino)
- Database-Independent Analysis of LC-MS/MS datasets with compareMS2 (Magnus Palmblad)
- Demonstration of a python package designed for quality control (Niveda Sundararaman)
- Solve the protein puzzle - Use the Human Protein Atlas to understand the different pieces of the human proteome (Cecilia Lindskog)

Thursday:

- Overview and challenges in the lipidomic data analysis (Nils Hoffman)
- Deep learning enabled software solutions for discovery proteomics, immunopectidomics, and glycoproteomics (BSI)
- How to analyse single cell proteomics data with the scp-package (Laurent Gatto)
- Training and transfer learning deep learning models to predict peptide property values with AlphaPeptDeep (Wen Feng-Zeng)
- BRAIN, Pointless4Peptides, MIND and QCQuan (Dirk Valkenborg)

On Wednesday, a special session on Scientific Communication was hosted by Ben Orsburn wherein he explained how he started a blog on proteomics <https://proteomicsnews.blogspot.com/> and how it affected his career in industry. Afterwards, Ben Orsburn and Ben Neely hosted the first live episode of the “The Proteomics Show” with Micha Birklbauer as their guest.

The program was completed by 56 posters that were presented on Tuesday evening in a long and active poster session. Among these, 8 were selected for flash talks presented to the entire

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audience. During the poster session, we also introduced our “Timeline” game, which was a huge success and attracted a lot of attention from the community. Overall –and despite the difficulties associated with the hotel (not enough rooms for everyone)– the EuBIC-MS Winter School was a success. Overall satisfaction of the participants as reported through the post-event feedback form (39 out of 116 participants responded).

More information can be found here: <https://eubic-ms.org/events/2024-winter-school/>

Developers Meetings

The EuBIC-MS Developers Meetings are dedicated to computer scientists and developers in the field of bioinformatics applied to MS, where they can discuss and work together in an open and constructive spirit. The program is split between keynote lectures and multiple hackathon sessions where the participants develop bioinformatics tools and resources that address outstanding needs in the bioinformatics community and among biologists using MS data. Our first Developers Meeting took place in Ghent, Belgium in 2018 (Willems et al., 2018), the second took place in Nyborg (Denmark) in 2020 (Ashwood et al., 2020), and the third took place in 2023 from 15-20 January at the ETH meeting platform Congressi Stefano Franscini (CSF), Monte Verità, Ticino (Switzerland). A proposition has already been made to organize the Developers Meeting 2025 in the Netherlands or Sweden.

Community achievements and outreach

The SDRF-Proteomics format

The EuBIC-MS community collaborated with the Human Proteome Organization Proteomics Standards Initiative (HUPO-PSI) (Deutsch et al., 2017) to define a standard for metadata annotation of proteomics data: The Sample and Data Relationship Format for Proteomics file format (SDRF-Proteomics) (Dai et al., 2021). This should increase data interpretability, reproducibility, as well as facilitating data reuse in an automated fashion. Unfortunately, community-wide adoption has been slow due to its complexity and the lack of user-friendly annotation tools. To address this, the CompOmics group developed this year a browser-based, intuitive application for metadata annotation called lesSDRF (Claeys et al., 2023). This tool allows users with minimal knowledge on SDRF-Proteomics to correctly annotate their data prior to submission to a public repository. In order to achieve this community-wide adoption of

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SDRF-Proteomics, several tool developers have been approached to include SDRF compatibility in their tools. So far the following tools have joined forces: FragPipe, MSStats, MSqRob, Mascot and PeptideShaker.

In addition to this, we want to further develop SDRF-Proteomics to keep up with the ever-evolving field of proteomics. Currently, there are active discussions ongoing on inclusion of metaproteomics, metabolomics and DIA acquisition on the dedicated GitHub repository (github.com/bigbio/proteomics-sample-metadata). This will expand to single-cell proteomics and immunopectidomics so that comprehensive metadata annotation for niche topics in proteomics can be assured.

Rusteomics

The aim of Rusteomics is to build, in the Rust programming language, a collaborative and community-driven toolbox to efficiently process MS data. The 2023 edition of the EuBIC-MS Developers Meeting provided an opportunity to establish a task force for the Rusteomics project. The hackathon jump started the project with the development of an application that creates a spectral library (NIST MSP format) from a peak list file (MGF format) and peptide-spectrum match (PSM) data (psm_utils TSV format). The prototyped tool reannotates MS/MS spectra based on matched peptide sequences. This Rust implementation proved to be significantly faster than its Python equivalent, with potential for further improvements. Initial attempts to integrate the C# ThermoRawFileReader were inconclusive. Organizational groundwork was laid, including setting up software modules, a GitHub organization, and essential project files. With a new team of maintainers, Rusteomics is poised for future expansion. However, despite the strong interest of project contributors, the project hasn't made significant strides over the past year. The project management needs enhancements to facilitate regular meetings and outline the necessary tasks to be performed.

ProteoBench

ProteoBench is a community-curated benchmark platform for comparing MS data analysis workflows. It started early 2023, and grew to gather more than thirty members on its dedicated EuBIC-MS slack channel. In February, we gathered for a very productive 5-days workshop organized by Matthias Mattanovich and Holda Angho in Copenhagen (Denmark). It was

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hosted in the University of Copenhagen and fully funded by the Danish Data Science Academy (DDSA). We also met for two days in September, just before the EuBIC-MS organizational meeting. During the 2024 EuBIC-MS Winter School, Robbin presented Proteobench in a 5-min flash talk and was followed with a poster presentation where several members jointly presented the project. On the 27th of February, Proteobench was officially launched during a webinar wherein we presented the platform.

In September, Robbin Bouwmeester and Bart Van Puyvelde hosted a hackathon in Ghent, where a lot of progress was made.

ProteoBench has a very active github repository (github.com/Proteobench/ProteoBench) that contains a soon-to-be working version of our first benchmark module focused on the comparison of workflows dedicated to the quantitative analysis of DDA data. It also contains detailed documentation on how to contribute to the project, how to use the platform, and how to propose new modules. We plan to present the project at the next EuBIC-MS Winter School and have an official launch for beta testing soon after in a webinar.

EuBIC-MS seminars 2024

As part of our annual organizational meeting, we had a seminar on Computational Mass Spectrometry. The seminar was held at Centre for Genomic Regulation (CRG) in Barcelona, Spain on the 26th of September. EuBIC-MS members and local researchers shared their work with 12 exciting talks. More information on the following link:

(<https://eubic-ms.org/events/eubic-ms-seminar-on-computational-mass-spectrometry-2024-barcelona/>).

Our website and resources

In partnership with the EuPA Educational Committee (EC), the Proteomics Academy web resource (www.proteomics-academy.org) has become the central communication portal for EuBIC-MS since its creation in 2015. To increase the visibility of EuBIC-MS itself, we also maintain a website dedicated to the initiative at eubic-ms.org. There, people find all information

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regarding past and upcoming EuBIC-MS activities, including this annual report. We recently added information on how we operate and how to acknowledge EuBIC-MS in publications.

This year, we officially transferred the proteomics Q&A page that was a google group to a dedicated discussion page on the EuBIC-MS github (github.com/orgs/EuBIC/discussions/categories/q-a).

How to get involved?

The EuBIC-MS community is open to everybody working in computational proteomics and mass spectrometry. You can join us by using the “become a member” [form](#) available on the eubic-ms.org website. An automatically generated email will be sent out to invite you to join the [EuBIC-MS Slack workspace](#), which is our extensively used platform to interact with other members. Additionally, you can follow and retweet our latest activities from our Bluesky account ([@eubic-ms.org](https://bsky.app/profile/@eubic-ms.org)). By joining us, you will be part of a community effort, and thus contribute to the sustainability of these scientific activities, which we expect to foster the development of the computational MS field.

Acknowledgments

The EuBIC-MS organizing committee would like to thank the European Proteomics Association and the National Societies for their continuous support and funding since the creation of the initiative. We also want to thank all the sponsors that support our workshops and events. And of course, we would like to underscore the remarkable work accomplished by all organizers of the previous Winter Schools, and Developers Meetings.

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